NAME:

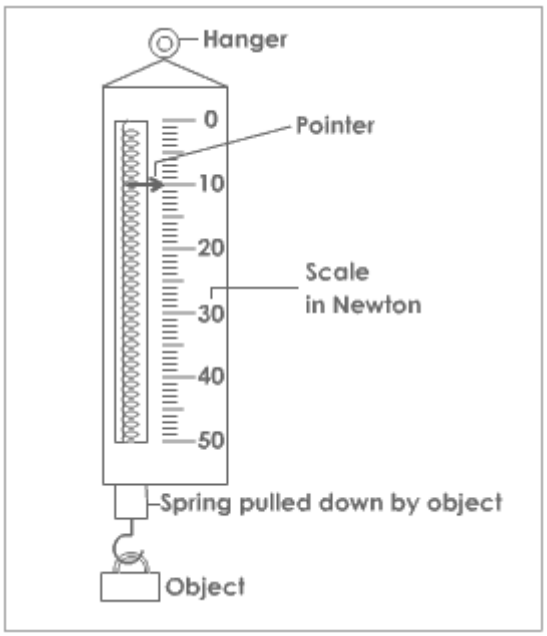

force
forces
Draw
weights
one
Trial
diagram
two
weight
object
balance
surface
piece
along
objects
use
hair
shape
paper
Use
Place
height
ruler
times
Two
directions
variable
newtons
Weight
time
first
arrows
resistance
Forces
showing
BONUS
far
cart
Balancing
Weights
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container
speed
four
measure
friction
unbalanced
diagrams
ball
show
put
2m
much
different
loose
spring
falling
slows
names
designed
happens
repulsion
grams
Im
side
amount
returned
Rub
Across
large
direction
Drop
causes
investigation
pieces
change
Add
gravity
centre
Overall
Ping
repulsive
falls
gravitational
bounce
Write
scrunched
make
set
correct
contact
Three
educated
touching
weigh
Flat
sizes
opposes
like
dependent
attraction
measured
long
Scrunched
guess
float
string
ground
Magnets
Explain
tennis
Put
constant
attractive
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proportionate
enables
independent
takes
reduce
motion
Mass
test
near
marble
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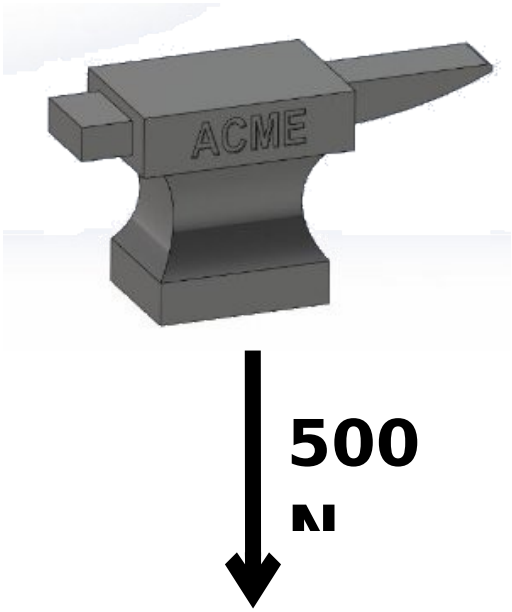
Spring-balance or Forcemeter

Weight is a force. Forces are measured in **newtons**. Measure the weight of these objects using the spring-balance. Draw each of the objects and the force in **newtons** like in the example below.

Is Weight a contact or noncontact force? _____

BONUS: If 1N = 100grams how much does each object weigh in grams?





Name of Object:	Name of Object:	Name of Object:
Weight in grams:	Weight in grams:	Weight in grams:

Magnets



Put the magnets together

1. Do the magnets have to be touching in order to experience the force?

   _____

2. How can you change the forces between two magnets from attractive to repulsive?

   _____

3. What is the force called?

   _____

4. Draw **two diagrams** to show the forces between magnets in the two cases or attraction and repulsion.

Attraction	Repulsion

Balancing Weights

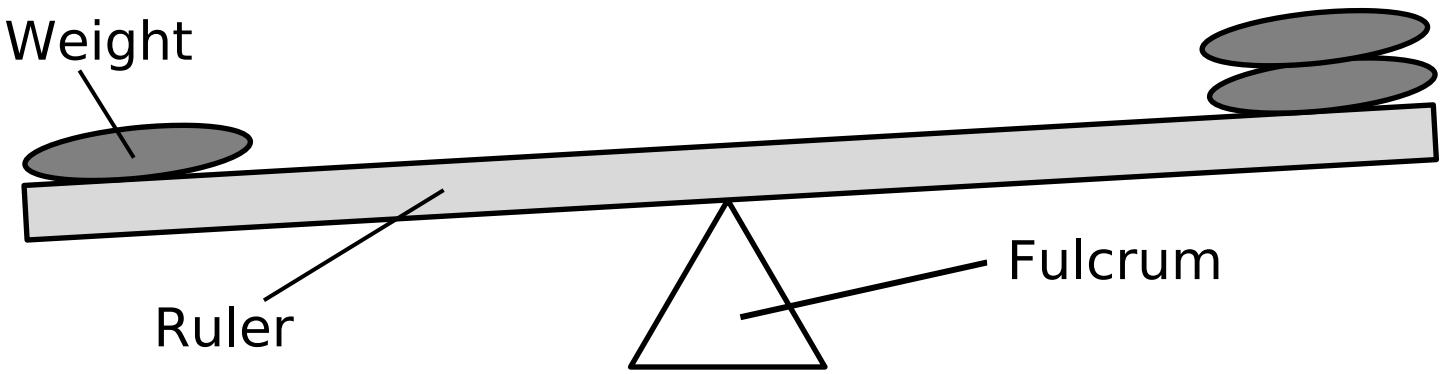
Set up the ruler with the fulcrum (balancing point) in the very centre. Use blu-tac to attach two weights to one end. How far along do you have to place **one** weight on the other side to balance the ruler? _____

If you three weights on one side how far along do you have to place the **one** weight? _____

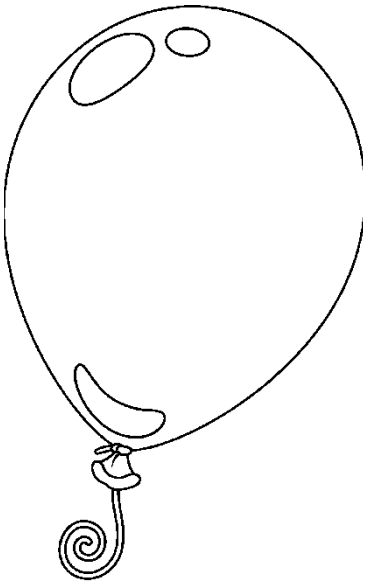
Hypothesise (make an educated guess) where you would put **one** weight to balance four weights:

_____

Bonus: If you have time test this hypothesis. Is it supported (that is, were you correct)? _____. Use **two** weights to balance the four weights.



Balloon



Rub the balloon on your clothes and then put the balloon near some loose hair.

1. Draw a diagram of what happens.
2. What causes the hair to stick to the balloon?

   _____

3. Add the names of the forces acting to the diagram and use arrows to show the directions of the forces.

Tennis Ball Drop

Drop the tennis ball from a 2m height and measure the height of the first bounce. Do this 3 times and repeat at a 1m height.

Height	Trial 1	Trial 2	Trial 3
2m			
1m			

In this investigation what is the independent variable? _____

What is the dependent variable? _____

What are some of the controlled variables? _____

_____

Why isn't all the gravitational force returned in the rebound?

_____

Draw a diagram of the investigation below:

Falling Paper

You should have one piece of paper that is scrunched up into a ball and one piece that is flat.

Drop the pieces of paper from shoulder height and time how long it takes to fall to the ground.

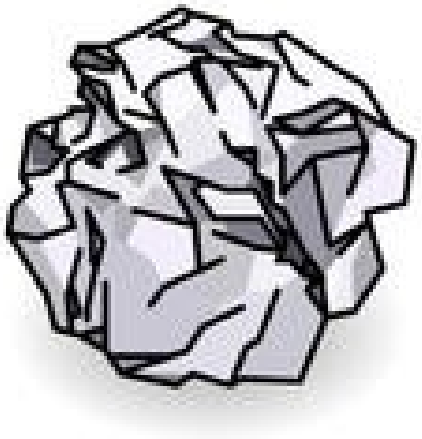
1. Write the times below:

	Trial One	Trial Two	Trial Three
Scrunched			
Flat			

2. What forces are acting on the paper as it falls? _____ and _____.
3. Add these forces to the diagrams, and use arrows to show the directions and proportionate size.
4. Explain why there is a difference in the times of the two pieces of paper.

_____

_____





Will it Float?

In the first container of water place the ping pong ball. In the second place the marble. Which one floats? _____ Why? _____.

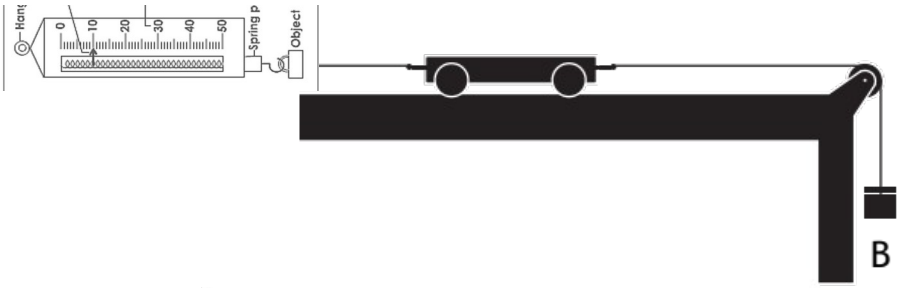
Draw a diagram for each container showing the forces acting (their direction and their relative sizes).

Keywords you should use in your diagram: gravity, upthrust, resistance

Overall are the forces balanced or unbalanced? _____

How Strong is the Force?

Use the spring balances to measure the force the weights exert on the cart.

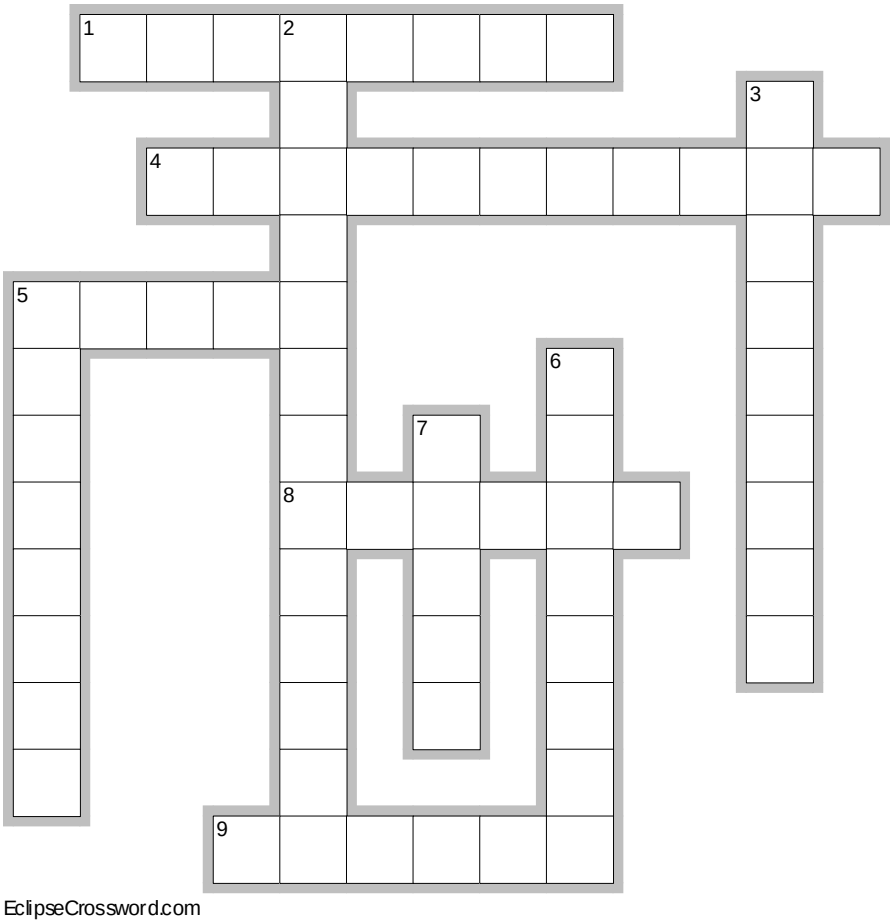


	Newtons Measured	
Mass added (g)	Trial One	Trial Two
50 g		
100 g		
150 g		
200g		

Can you see a pattern? _____

Draw a diagram below showing the different forces acting on the cart, string and weights.

Forces



Across

1. The forces when an object is stationary or at constant speed
4. A shape designed to reduce air resistance
5. Can change an object's speed, shape or direction
8. A surface with very little friction
9. The force on an object due to gravity

Down

2. A force which slows objects falling through air
3. The winning force if the forces are unbalanced
5. A force which opposes motion
6. A force which enables an object to float
7. A surface which causes a large amount of friction